Session 6 Solutions

Backpropagation & Chain Rule | Instructor copy

Question 1: MSE by Hand

Sample	True y	Predicted $\hat{y}$	Squared Error
1	2.0	2.5	0.25
2	4.0	3.0	1.00
3	1.0	1.5	0.25

$$\text{MSE} = \frac{0.25 + 1.00 + 0.25}{3} = \frac{1.50}{3} = 0.5$$

Question 2: Why Squaring?

Without squaring, positive and negative errors cancel each other out a model that over-predicts by 5 and under-predicts by 5 would appear to have zero error. Squaring makes all contributions positive and penalises large errors more heavily (quadratic growth), which drives the optimiser to reduce the worst predictions first.

Question 3: Derivative of a Loss

Let $u = 4 - 2w$, so $L = u^2$. By the chain rule:

$$\frac{dL}{dw} = 2u \cdot \frac{du}{dw} = 2(4 - 2w)(-2)$$

$$\frac{dL}{dw} = -4(4 - 2w) = 8w - 16$$

Question 4: Gradient Direction

We should **increase** w. The gradient descent update rule is:

$$w \leftarrow w - \eta \frac{dL}{dw}$$

A negative gradient means subtracting a negative number, i.e. w increases. Intuitively, the loss curve slopes downward to the right at $w = 0$, so moving right (increasing w) reduces the loss.

Question 5: Manual Gradient Descent

$\frac{dL}{dw} = 2w - 6$, true minimum at $w^* = 3$ (where $L(3) = 1$).

Step	w	$L(w)$	$\frac{dL}{dw}$	New w
0	0.000	10.000	-6.000	$0 - 0.2 \times (-6) = 1.200$
1	1.200	4.840	-3.600	$1.2 - 0.2 \times (-3.6) = 1.920$

w is converging toward 3.0 ✓

Question 6: Loss Landscape

A **local minimum** is a point where the loss is lower than all nearby points, but not necessarily the lowest overall. A **global minimum** is the single lowest point on the entire loss surface.
For **linear models with MSE loss**, the loss surface is a convex paraboloid (a smooth bowl), so there is exactly one minimum, which is both local and global. Non-linear models such as MLPs can have multiple local minima and saddle points.

Question 7 : Partial Derivatives

Let $u = 2 - w_1 - 3w_2$, so $L = u^2$.

$$\frac{\partial L}{\partial w_1} = 2u \cdot (-1) = -2(2 - w_1 - 3w_2)$$
$$\frac{\partial L}{\partial w_2} = 2u \cdot (-3) = -6(2 - w_1 - 3w_2)$$

At $(w_1, w_2) = (1, 0)$: $u = 2 - 1 - 0 = 1$.

$$\nabla L\big|_{(1,0)} = \begin{bmatrix} -2 \\ -6 \end{bmatrix}$$

Question 8 : SGD Variants

Variant	Batch size	Gradient estimate	Typical use case
Full-batch GD	N	Exact	Small datasets; convex problems
Mini-batch SGD	$B \ll N$	Approximate, low noise	Standard practice for deep learning
Stochastic GD	1	Very noisy	Online learning; very large datasets

Question 9 : Diverging Loss

The most likely cause is a **learning rate that is too large**. With an oversized η, each update overshoots the minimum the weights jump past it and land on the other side at a higher loss value, which compounds with each step.
Fix: reduce the learning rate (e.g. divide by 10) and restart training. A useful diagnostic is to plot the loss curve on a log scale; divergence appears as a rapid upward trend rather than a plateau.

Question 10 : Bridge to Neural Networks

The four-step recipe (forward pass → loss → gradients → weight update) is identical. The key difference is **step 3 computing gradients for hidden layers**. In linear regression the gradient is a single, direct derivative. In an MLP, weights in early layers affect the loss *indirectly*, through the hidden activations:

$$\frac{\partial L}{\partial W^{(1)}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} \cdot \frac{\partial h}{\partial W^{(1)}}$$

This requires the **chain rule** applied layer by layer, which is exactly what **backpropagation** (Session 6) implements efficiently.

Question 11 : Chain Rule Symbolic Derivative

Let $z = 3x - 2$, so $y = \sigma(z)$. By the chain rule:

$$\frac{dy}{dx} = \frac{d\sigma}{dz} \cdot \frac{dz}{dx} = \sigma(z)\big(1 - \sigma(z)\big) \cdot 3$$

$$\frac{dy}{dx} = 3\,\sigma(3x-2)\big(1 - \sigma(3x-2)\big)$$

Question 12 : Chain Rule Numerical

Let $u = 2x + 1$, so $y = u^3$. By the chain rule:

$$\frac{dy}{dx} = 3u^2 \cdot \frac{du}{dx} = 3(2x+1)^2 \cdot 2 = 6(2x+1)^2$$

At $x = 1$: $u = 3$, so:

$$\left.\frac{dy}{dx}\right|_{x=1} = 6 \times 3^2 = 54$$

Question 13: Forward Pass by Hand

Parameters: $w = 0.8$, $b = -0.3$, $x = 1.5$, $y = 1$.

Step	Formula	Expression	Value
1	$z = wx + b$	$0.8 \times 1.5 - 0.3$	0.9
2	$\hat{y} = \sigma(z)$	$\sigma(0.9) = \dfrac{1}{1+e^{-0.9}}$	≈ 0.7109
3	$e = \hat{y} - y$	$0.7109 - 1$	≈ -0.2891
4	$L = e^2$	$(-0.2891)^2$	≈ 0.0836

$z = 0.9, \quad \hat{y} \approx 0.7109, \quad L \approx 0.0836$

Question 14: Backward Pass by Hand

Using the values from Question 13: $e \approx -0.2891$, $\hat{y} \approx 0.7109$, $z = 0.9$, $x = 1.5$.

Step	Gradient	Value
1	$\dfrac{\partial L}{\partial e} = 2e$	$2 \times (-0.2891) \approx -0.5782$
2	$\dfrac{\partial L}{\partial \hat{y}} = \dfrac{\partial L}{\partial e} \cdot 1$	≈ -0.5782
3	$\dfrac{\partial L}{\partial z} = \dfrac{\partial L}{\partial \hat{y}} \cdot \sigma(z)(1-\sigma(z))$	$-0.5782 \times 0.7109 \times 0.2891 \approx -0.1189$
4	$\dfrac{\partial L}{\partial w} = \dfrac{\partial L}{\partial z} \cdot x$	$-0.1189 \times 1.5 \approx -0.1784$
5	$\dfrac{\partial L}{\partial b} = \dfrac{\partial L}{\partial z} \cdot 1$	≈ -0.1189

Both gradients are **negative**: gradient descent will *increase* w and b, pushing $\hat{y}$ closer to 1. ✓

Weight update with $\eta = 1.0$:

$$w \leftarrow 0.8 - 1.0 \times (-0.1784) = 0.9784$$
$$b \leftarrow -0.3 - 1.0 \times (-0.1189) = -0.1811$$

$w_{\text{new}} \approx 0.9784, \quad b_{\text{new}} \approx -0.1811$

Question 15 : Error Signal δ

Given $\delta^{(2)} = -0.20,\ W^{(2)} = [0.6,\ -0.4]^T,\ z^{(1)} = [0.6,\ -0.3]^T$.

(a) Pull the error back through the weights:

$$(W^{(2)})^T\,\delta^{(2)} = \begin{bmatrix} 0.6 \\ -0.4 \end{bmatrix} \times (-0.20) = \begin{bmatrix} -0.120 \\ 0.080 \end{bmatrix}$$

(b) Sigmoid derivative at each hidden pre-activation:

$$\sigma(0.6) \approx 0.6457, \qquad \sigma'(0.6) = 0.6457 \times 0.3543 \approx 0.2288$$
$$\sigma(-0.3) \approx 0.4256, \qquad \sigma'(-0.3) = 0.4256 \times 0.5744 \approx 0.2445$$

$$\sigma'(z^{(1)}) \approx \begin{bmatrix} 0.2288 \\ 0.2445 \end{bmatrix}$$

(c) Element-wise product:

$$\delta^{(1)} = \begin{bmatrix} -0.120 \\ 0.080 \end{bmatrix} \odot \begin{bmatrix} 0.2288 \\ 0.2445 \end{bmatrix}$$

$$\boxed{\delta^{(1)} \approx \begin{bmatrix} -0.0275 \\ 0.0196 \end{bmatrix}}$$

Question 16 : Vanishing Gradients Conceptual

(a) At each layer the backward-propagating gradient is multiplied by $\sigma'(z) \le 0.25$. After L sigmoid layers the gradient is scaled by a factor of at most 0.25^L, which shrinks *exponentially* toward zero. Early-layer weights therefore receive gradients so small that they effectively stop learning this is the **vanishing gradient problem**.

(b) The **Rectified Linear Unit (ReLU)** was introduced to address this:

$$\mathrm{ReLU}(z) = \max(0,\, z), \qquad \mathrm{ReLU}'(z) = \begin{cases} 1 & z > 0 \\ 0 & z \le 0 \end{cases}$$

When a neuron is active ($z > 0$), the derivative is 1, so the gradient passes through unchanged and does not shrink as it flows backward through ReLU layers.